# Unit 5.4 The Fundamental theorem of calculus



If $f$ is continuous on $[a,b]$ then

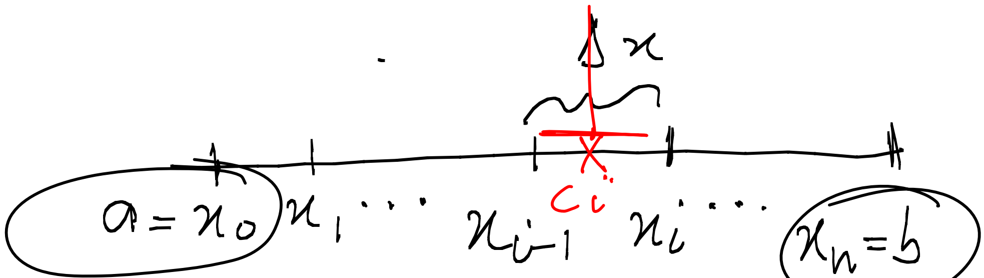
$$\int_a^b f(x)\,dx = F(b) - F(a)$$

where [illegible] is any anti-derivative of $f$,
that is [illegible]

__Proof:__

Divide $[a,b]$ into [illegible] subintervals of length $\Delta x = \dfrac{b-a}{n}$.

Let $a = x_0 < x_1 < \cdots\cdots < x_{n-1} < x_n = b$
be the endpoints of the interval.




We use the MVT on $[x_{i-1}, x_i]$

$F$ is db on $[a,b]$ $\quad (F' = f)$
So $F$ is db on $[x_{i-1}, x_i]$

$F$ is cont on $[a,b]$ $\quad$ ($F$ is db)
So $F$ is cont on $[x_{i-1}, x_i]$

So there exists a $c_i \in (x_{i-1}, x_i)$
such that $F'(c_i) = \dfrac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}}$

Pick $c_i$ as the sample point in $[x_{i-1}, x_i]$

Then

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} \underbrace{f(c_i)}_{\text{Sample point}} \cdot \Delta x \qquad \Big| \; F' = f$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} F'(c_i)\,\Delta x$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{F(x_i)-F(x_{i-1})}{\cancel{x_i - x_{i-1}}} \cdot \cancel{(x_i - x_{i-1})}$$

$$= \lim_{n\to\infty} \Big(F(x_1) - F(x_0) + F(x_2) - \cancel{F(x_1)} + \cdots\cdots + F(x_n) - F(x_{n-1})\Big]$$

$$= \lim_{n\to\infty} \Big[F(x_n) - F(x_0)\Big]$$

$$= \lim_{n\to\infty} \big(F(b) - F(a)\big).$$

$$= F(b) - F(a).$$

## Examples


1. Find $\displaystyle\int_0^{\pi/2} \underbrace{\sin x}_{f} \, dx$

$$= -\cos x \Big]_0^{\pi/2} = -\cos\tfrac{\pi}{2} - (-\cos 0)$$
$$= 1.$$

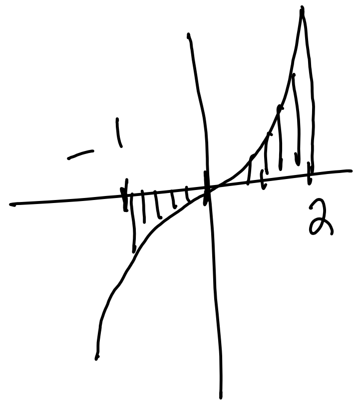
2. Find $\displaystyle\int_{-2}^{2} (x^2+1)\,dx$

$$= \left(\tfrac{1}{3}x^3 + x\right)\Big]_{-2}^{2} = \tfrac{8}{3} + 2 - \left(-\tfrac{8}{3} - 2\right)$$
$$= \tfrac{16}{3} + 4 = \tfrac{28}{3}.$$

## Area problems

sketch.

1. Find the area enclosed between $y = x^3$ and the $x$-axis between $x = -1$ and $x = 2$.

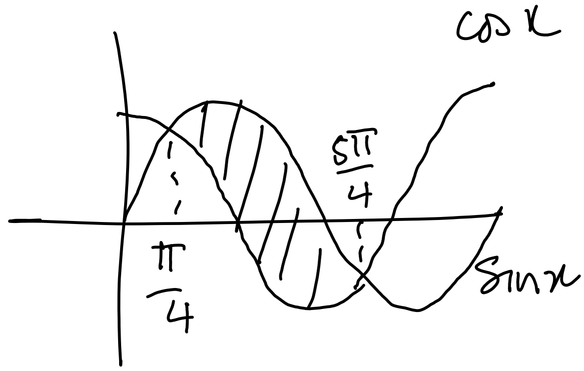



Area: $\displaystyle -\int_{-1}^{0} x^3\,dx + \int_0^2 x^3\,dx$

because $f$ is neg and the area has to be pos.

$$= -\tfrac{1}{4}x^4\Big]_{-1}^{0} + \tfrac{1}{4}x^4\Big]_0^2 = \tfrac{1}{4} + 4 = \tfrac{17}{4}$$

2. Find the area enclosed between $y = \sin x$ and $y = \cos x$ between $x = \pi/4$ and $x = \frac{5\pi}{4}$




Area: $\displaystyle\int_{\pi/4}^{5\pi/4} (\sin x - \cos x)\,dx$

$= (-\cos x - \sin x)\Big|_{\pi/4}^{5\pi/4}$

$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right)$

$= \frac{4}{\sqrt{2}} = 2\sqrt{2}.$

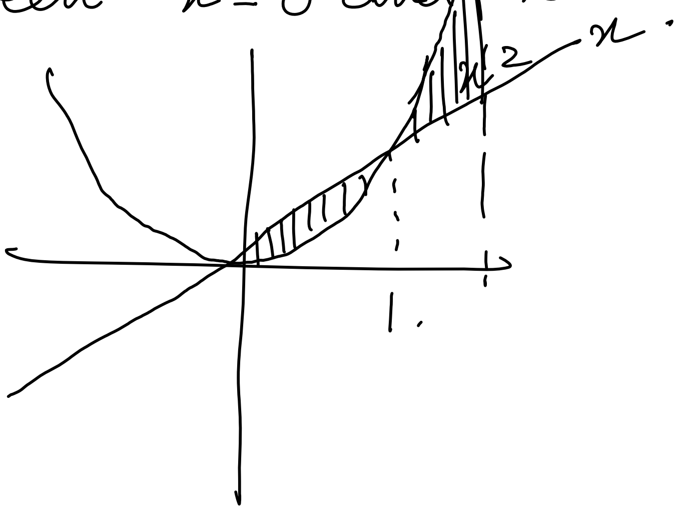
Area between two curves

$\int (\text{top} - \text{bottom}).$

NB.

3 Find the area enclosed between $y = x^2$ and $y = x$ between $x = 0$ and $x = 2$.

sketch.




$\displaystyle\int_0^1 (x - x^2)\,dx + \int_1^2 (x^2 - x)\,dx$